

Surface Metrics

HES 505 Fall 2025: Session 17

Matt Williamson

Objectives

- **Describe** the importance of measuring recurrent landscape patterns
- **Distinguish** between patch, class, and landscape-level metrics
- **Implement** patch metric calculations using the **landscapemetrics** package for **R**

**What do we mean by
landscape?**

Definining landscapes



Organism

A single individual



Population

Individuals of the same species in the same habitat



Community

Populations of several species in one habitat



Ecosystem

A habitat's community and its abiotic factors



Landscape

The patchwork of ecosystems across a large area



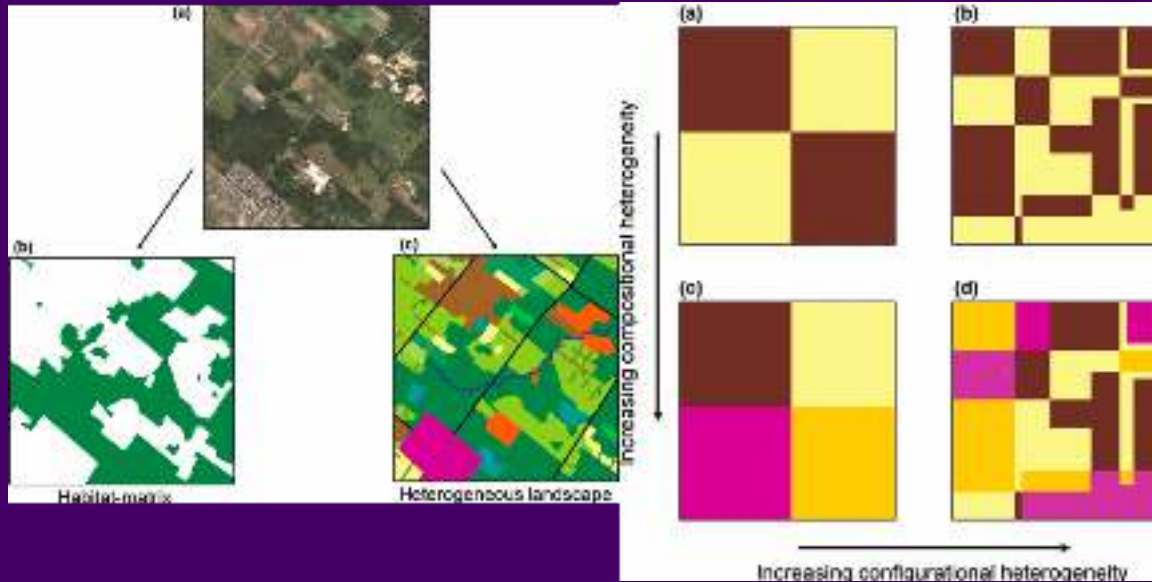
Biosphere

All landscapes inhabited by living organisms on Earth

Core Concepts

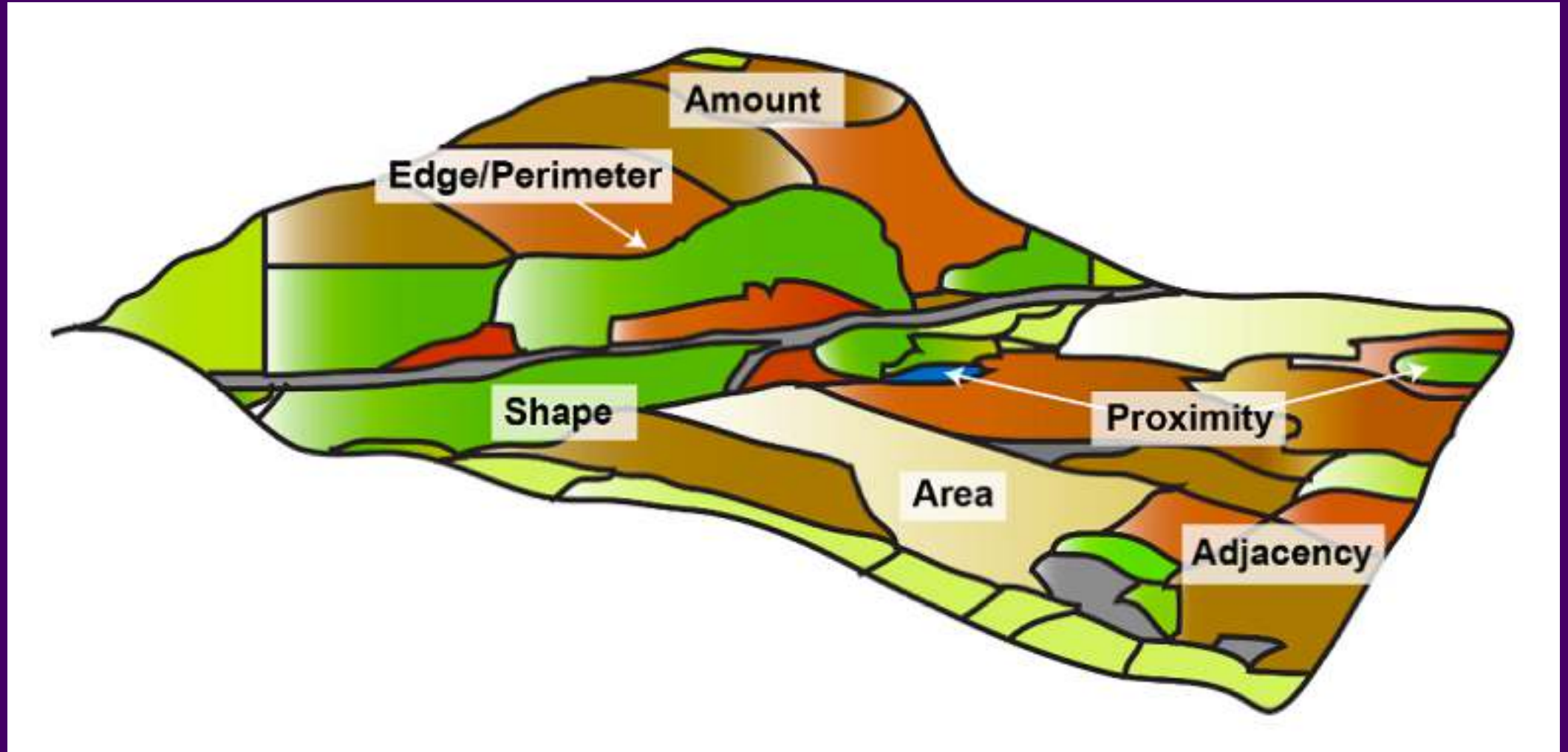
- Heterogeneity is a defining feature of a landscape
- Landscapes can be defined and studied at any scale
- Patterns and processes occur across scales
- Not limited to terrestrial systems

Heterogeneity and Dynamics

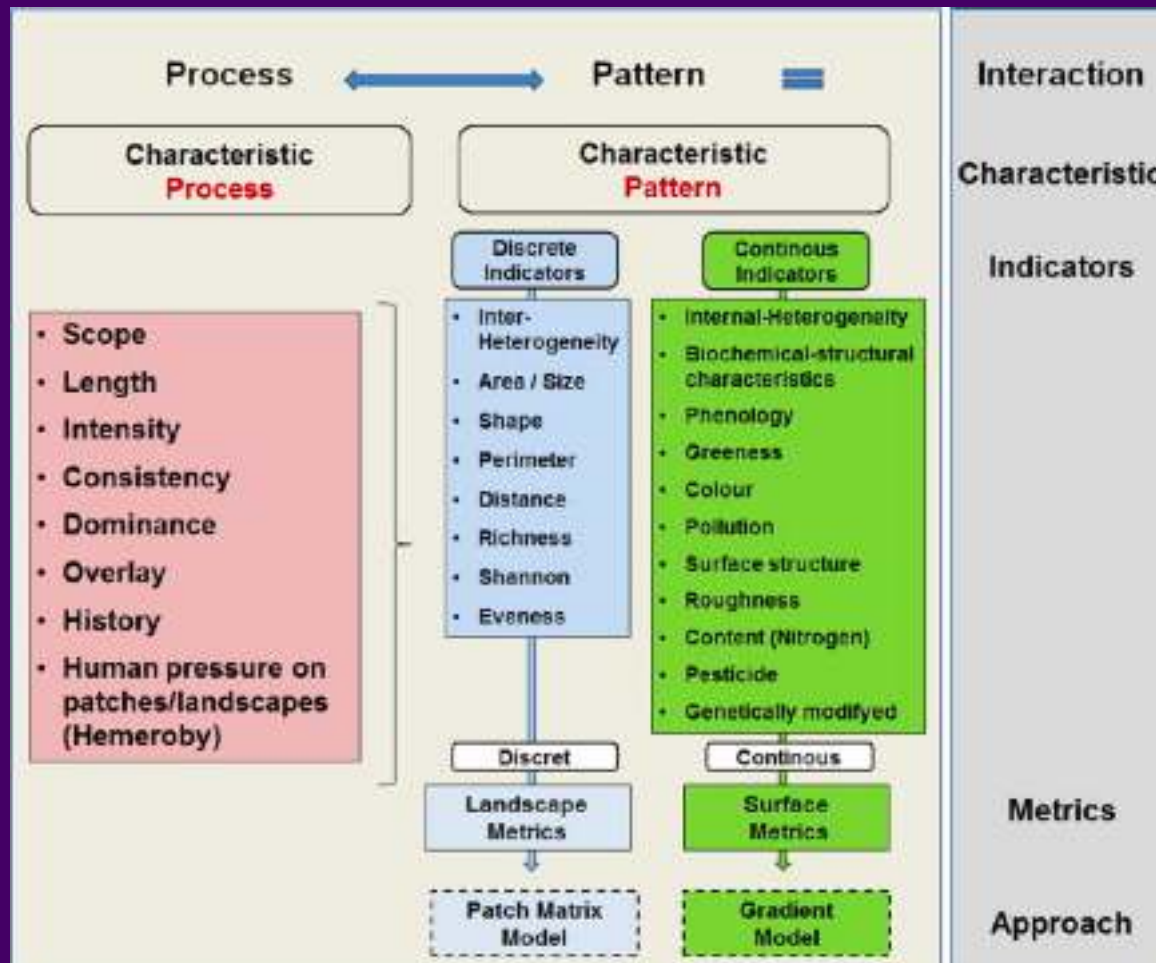


- Composition
- Matrix vs. patch-mosaic vs. gradients
- Community composition
- Turnover and invasion

Measuring Patches



Measuring Patches



Measuring Patches

- **Patch** metrics: describe attributes of each patch
- **Class** metrics: describe the distribution of patches within each category
- **Landscape** metrics: describe the composition of the landscape based on patches and categories

Scale Matters!!!

Most Metrics Are Correlated

- Combinations are possible (PCA - Wednesday)
- **Strength** - The average variance in configuration explained by a metric/combination
- **Consistency** - Do the metrics/ combinations mean the same thing in different patches/landscapes
- **Universality** - The percentage of regions where the metric/combination plays a role

Why Do We Care

- Describe the landscape mosaic
- As a foundation for varying rates in point process models
- As predictors for other models
- As an objective for planning

